

Chapter 5 Warm-Up: Python **for** and **while** Loops

Prof. Jordan C. Hanson

1 Student Data

1. A list can contain tuples representing individual students. Each tuple below contains a student's name, student ID, and math placement score:

```
students = [  
    ("Maya", 20681001, 5),  
    ("Noah", 20681002, 1),  
    ("Olivia", 20681003, 3),  
    ("Liam", 20681004, 4),  
    ("Sophia", 20681005, 2)  
]
```

Each student therefore has the form

```
(name, student_id, score)
```

Try

```
print(students[0])  
print(students[0][0])  
print(students[0][1])  
print(students[0][2])
```

Identify which expression produces the student's name, ID, and math score.

2 for Loops

1. A **for** loop performs the same operation on each element of a collection. Enter

```
for student in students:  
    print(student)
```

The variable **student** refers to one tuple during each pass through the loop.

- Modify the loop to extract all three values:

```
for student in students:  
    name = student[0]  
    student_id = student[1]  
    score = student[2]  
    print(name, student_id, score)
```

- Add conditional logic inside the loop so that each student's recommended course is printed:

```
if score >= 4:  
    print("Calculus")  
elif score >= 2:  
    print("Trigonometry")  
else:  
    print("Algebra")
```

3 while Loops

1. A **while** loop repeats while a Boolean condition remains **True**. An index can be used to select one student at a time:

```
i = 0  
  
while i < 5:  
    print(students[i])  
    i = i + 1
```

- What value does **i** have during the first pass through the loop? During the last pass?
- What would happen if

```
i = i + 1
```


were omitted?
- Modify the loop so that it prints only each student's name and student ID.

4 Warm-Up Exercise

1. Create the following initially empty course structures:

```
algebra = ["Algebra", []]  
trig = ["Trigonometry", []]  
calculus = ["Calculus", []]
```

```
math_courses = (  
    algebra,  
    trig,  
    calculus  
)
```

Use a **for** loop to examine every student in **students**. Use an **if-elif-else** statement to determine the correct course. Instead of adding only the student's name, add a tuple containing the student's name and ID:

```
(name, student_id)
```

For example,

```
calculus[1].append(  
    (name, student_id)  
)
```

When the loop is finished, print

```
print(math_courses)
```

and verify that

- scores 0–1 are placed in Algebra,
- scores 2–3 are placed in Trigonometry,
- scores 4–5 are placed in Calculus, and
- every roster contains both student names and IDs.